Executive Summary: For our last week the team finished everything. Our documentation on both the hardware and software side has been completed and the additional documentation has been completed and committed to the github. This week we also finished our Final demo presentation and practiced it multiple times in order to get a better understanding for our final presentation in person on Monday 5/4. The team also created a case for the WAD and although they ran into issues with the initial design they were able to get a working usable case on the WAD. Our team is ready and waiting to give our final presentation at which point all documentation will be handed over in the form of the presentation folders and the USB archives.

Task / Time Take / Completed Y/N

Hardware:

Cody C.

Progress:

- 4/26 –
    - Work on Presentation Slides (~1 Hours)
        - Changes made to the project:
            - I2S Microphone to MEMS Microphone
                - Everything is the same, omnidirectional, 59dB Noise to sound ratio
                - Main difference is the Sound Pressure Level (140dB vs 110dB)
        - Future Planning:
            - Smaller form factor
            - PCB implementation

        - High Decibel Microphone
    - Calculated Run time for Wearable Alert Device
    - Edit GitHub (~20 Minutes)
- 4/29 –
    - Team meeting (~2 Hours 30 Minutes)
        - Field data collection for GPS module
        - Talk about plans for Final Demonstration
        - Practiced Final Presentation
    - Update Symbol for Micro-Lipo Charger (~5 Minutes)
    - Update Schematics and Wiring Diagram (~5 Minutes)
    - Work on Hardware Design Document (~3 Hours)
        - Create descriptions for Arduino Nano ESP32, LED, and MEMS Microphone
        - Upload Schematics
        - Upload Block Diagrams
            - Visual of Microphone
            - Whole Black Diagram
    - Update GitHub (~10 Minutes)
- 5/2 –
    - Team Practice (~30 Minutes)

Total Hours Worked: ~ 7 Hours 40 Minutes

Future Work/Plan

- 5/4 - Final Demonstration at 5:30 pm

Task / Time Take / Completed Y/N

Hardware:

Ryan K.

Progress:

- 4/27 - Presentation Slides (~1hr 30min)
    - Worked on our Final Demo Presentation Slides for the upcoming team meeting
- 4/29 - Team Meeting/Presentation Practice (~2hr 30min)
    - Met with team to talk about upcoming needs with documentation
    - Gathered GPS data and made final edits to Final Demo Slides
    - Practiced the Final Demo with the team
- 4/29 - Documentation (~1hr 45min)
    - Started writing the Hardware Design Document
        - Filled out and gathered information for the battery and Micro-Lipo Charger
- 5/1 - Documentation (~1hr 30min)
    - Continued filling out the Hardware Design Document
        - Filled out and gathered information for the positioning module, and the boost converter
- 5/2 - Finalize Documentation (~20min)
    - Formatted and reviewed the Hardware Design Document and made minor edits where needed

Total Hours Worked: ~7hr 35min

Future Work/Plan:

- Present our Final Demo on 5/4 @ 5:20pm

Task / Time Take / Completed Y/N

Software:

Alex H.

Progress:

- 4/27 (3 h 52 min)
    - Created two gunshot detection videos for the Final Demo presentation, showing off results.
    - Computed accuracy of WAD for each gunshot detection video.
    - Shared video format with the team to receive their approval and feedback.
    - Inserted gunshot detection videos into the Final Demo presentation.
- 4/28 (1 h 41 min)
    - Formatted appendix and gunshot detection slides for the Final Demo presentation.
- 4/29 (2 h 21 min)
    - Worked with the team to obtain some new GPS tracking data to present at the Final Demo.
        - Obtained coordinates and altitude of the OU Engineering Quad fountain. Compared this data from the WAD with data from Google Maps and an altitude app.
    - Practiced Final Demo presentation with the team.
- 5/1 (5 h 37 min)

    - Finished the user manual for the Wearable Alert Device. Added images and steps for installing the nRF Connect app, as well as writing brief descriptions of each of the device's features.
- 5/2 (39 min)
    - Ran more practices with the team on the Final Demo presentation.
    - Discussed plans to merge the GitHub repository so that all project files are together.
    - Added slight modifications to the Final Demo presentation.
- 5/3 (44 min)
    - Double-checked that the README file, user manual, and code are all uploaded to the GitHub repository in their final versions.
    - Merged all branches in the GitHub repository to compile all project files together.

Total Hours Worked: 14 h 54 min

Future Work/Plan:

- Present project at the Final Demo presentation on Monday 5/4

Project Manager:

Jonas W.

Progress:

- 4/28 Finish my portion of final slides (1 Hours)
    - Create and finish my final budget slides and introduction slide
- 4/29 Team meeting (~2 Hours 30 Minutes)
    - Test the Position Module
    - Test the bluetooth range

- 4/30 Create the WAD Case (3 Hours)
    - Create the STL and Obj Files of the WAD Device’s case
    - Upload to the GIT
- 5/1 Send the Case to volunteer 3D printer (1 Hours)
    - Coordinate with the volunteer on the printing of the Case
    - Plan on pick up/drop off of case
- 5/3 Finish Documentation(2 Hours)
    - Finished the Final budget/BOM
    - Finished formatting of the HDD and the User Manual
    - Upload everything to GitHub
- 5/3 Build Case (1 Hours)
    - Got the case parts and finished assembling, ran into the issue of the device on the inside being a bit too long but we found a solution by adding lots of tape.

Total Hours Worked: 10 Hours 30 Minutes

Future Work/Plan:

Print and create the final demo folders

Format the USB archives

Present the final Demo!